

Notes: Gomes (AER, 2001)

Financing Investment

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1. Big Picture

Research question. The paper asks whether the common empirical finding that investment is sensitive to cash flow should be interpreted as evidence that firms face financing constraints. Gomes builds a structural model in which heterogeneous firms make entry, exit, investment, and financing decisions while external finance is costly. The paper then runs the same cash-flow-augmented investment regressions on simulated model data that researchers often run on real Compustat data.

Main answer. Cash-flow sensitivity is not a reliable diagnostic for financing constraints. In a correctly specified model, financing frictions affect firm value and therefore should already be reflected in Tobin's q . Cash flow becomes significant in reduced-form investment regressions when q is omitted, poorly measured, or when the linear regression is a bad approximation to a nonlinear investment rule.

Core mechanism. Firms differ in capital k and productivity z . Productive firms may want to invest aggressively, but external finance is costly. Some firms are constrained because the gap between desired investment and internal funds is not large enough to justify paying financing costs. Other firms are so productive, or so far from desired scale, that they pay the fixed/variable costs and raise outside funds. This selection generates rich cross-sectional patterns, but it does not imply that cash flow adds information once q is well measured.

2. Incremental Contribution

1. **Structural benchmark for a reduced-form debate.** The paper does not only criticize cash-flow sensitivity regressions verbally; it constructs a heterogeneous-firm equilibrium model with costly external finance, calibrates it, and asks what the standard regressions would find if the model were the data-generating process.
2. **Separates existence of constraints from empirical detectability.** The benchmark economy has many constrained firms, yet the cash-flow regressor often adds little explanatory power beyond q . This distinction is central: financing constraints can be economically important and still be poorly detected by the usual cash-flow coefficient.
3. **Shows non-necessity of constraints for cash-flow effects.** Significant cash-flow effects can appear even when the model has no financing constraints. This directly undermines the common inference from cash-flow sensitivity to financing frictions.
4. **Emphasizes measurement error in q .** The paper shows that when the econometrician observes a noisy proxy for investment fundamentals, cash flow can pick up omitted information about productivity and investment opportunities.
5. **Explains why size splits can be misleading.** Small firms are more likely to be constrained on average, but size is an imperfect proxy because productivity, exit risk, and external finance use vary across firms.

3. Model and Empirical Design

Model ingredients. The economy contains households, financial intermediaries, and heterogeneous firms. Firms face decreasing returns to scale, idiosyncratic productivity shocks, fixed operating costs, entry and exit, and costly external finance. A firm chooses whether to exit, how much capital to carry into the next period, and whether to finance investment internally or externally.

Financing costs. External finance is costly only when investment exceeds internal funds. The benchmark financing cost function has a fixed component and a variable component calibrated using equity flotation-cost evidence. This makes small issues especially expensive and generates an external-finance region only for firms with sufficiently high investment benefits.

Investment regression tested. The paper evaluates the standard investment equation

$$\frac{i_{it}}{k_{i,t-1}} = b_0 + b_1 q_{i,t-1} + b_2 \frac{\pi_{i,t-1}}{k_{i,t-1}} + f_t + d_i + \varepsilon_{it},$$

where i/k is investment scaled by lagged capital, q is average Tobin's q , π/k is cash flow scaled by lagged capital, f_t is a year effect, and d_i is a firm fixed effect. The paper compares this regression in Compustat data with the same regression run on artificial data generated by the model.

Interpretation of q . The key theoretical point is that q is a sufficient statistic for the value of an additional unit of capital only under strong conditions. If q is measured well in the model, it already contains information about financing frictions. If q is measured poorly, cash flow may appear significant because it is correlated with productivity and investment opportunities.

4. Main Results

- **Financing constraints are present in the model.** More than half of firms are constrained in the benchmark economy, and firms that use external finance account for a large share of investment.
- **Cash-flow sensitivity is weak evidence.** In the benchmark model, adding cash flow to the investment regression does not materially improve explanatory power beyond q .
- **Cash-flow effects can occur without frictions.** A model with no financing constraints can generate a large increase in regression fit when cash flow is added.
- **Subsample tests do not solve the problem.** Sorting by size or by the model's true constrained/unconstrained status does not make the cash-flow coefficient a clean test of financing frictions.
- **Measurement error matters.** When the econometrician uses a noisy or incorrect measure of q , cash flow can appear useful, even when the underlying reason is not a financing constraint.

5. Figure and Table Notes

Figure 1. Firm-Level Investment Shares

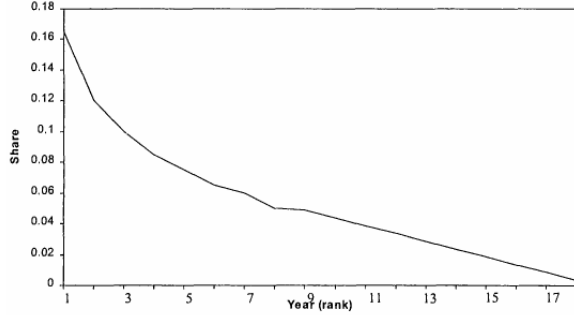


FIGURE 1. FIRM-LEVEL INVESTMENT SHARES

Interpretation. The figure documents lumpy investment at the plant level. The highest-ranked investment year accounts for a large share of cumulative investment, and investment is concentrated in a small number of years. This motivates the paper's departure from smooth representative-firm investment models and its use of heterogeneous firms with nonlinear investment policies.

Figure 3. Financing Costs

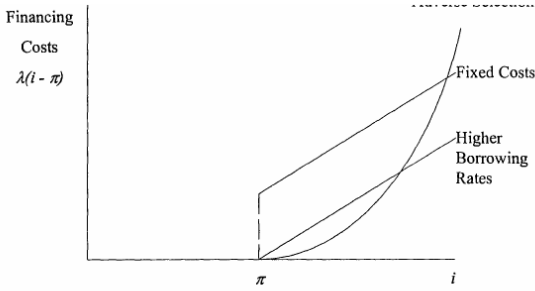


FIGURE 3. FINANCING COSTS

Interpretation. The figure illustrates possible shapes of the external-finance cost function. Financing costs are zero when internal funds cover investment and positive once the firm must raise outside capital. Fixed issuance costs, higher borrowing rates, and adverse-selection costs all generate a wedge between internal and external finance.

Figure 2. Timing of Events

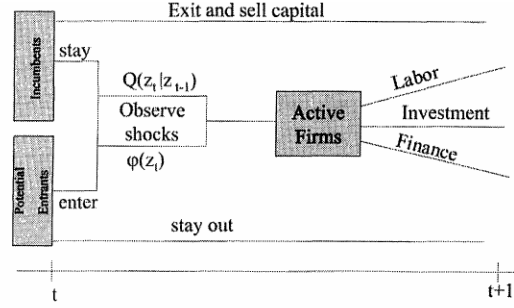


FIGURE 2. TIMING OF EVENTS

Interpretation. The timing diagram clarifies the dynamic problem. Incumbents first decide whether to stay or exit, potential entrants decide whether to enter, shocks are realized, and active firms choose labor, investment, and financing. The ordering matters because exit and entry create selection in the cross section of firms used for simulated regressions.

Figure 4. Costs of Issuing Securities

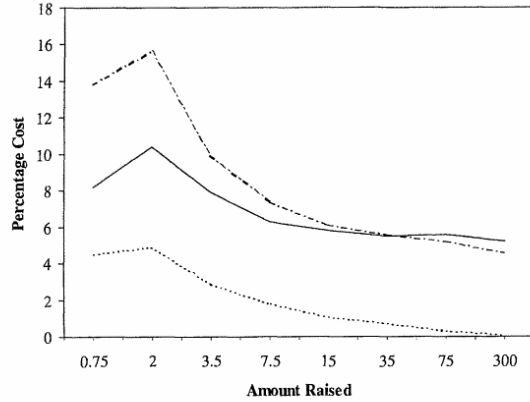


FIGURE 4. COSTS OF ISSUING SECURITIES

Note: The top line signifies underwriting; the middle line

Interpretation. The flotation-cost data show large economies of scale in external equity issuance. Small issues are especially expensive as a percentage of the amount raised. This empirical pattern justifies the model's fixed-plus-variable financing cost specification and explains why many firms avoid outside finance unless the investment opportunity is strong.

Table 1. Calibration

TABLE 1—CALIBRATION		
Parameter	Benchmark value	Empirical restriction
Technology		
α_k	0.3	Degree of returns to scale
α_l	0.65	Labor share
δ	0.145	Investment to capital ratio
Technology Shock		
ρ	0.762	Persistence in investment
σ	0.0352	Cross-section variance of investment
Financing Costs		
λ_0	0.08	Fixed flotation costs
λ_1	0.028	Unit flotation costs
Preferences		
β	1/1.065	Interest rate
H	0.6	Employment share

Interpretation. The benchmark parameters are chosen to match major moments: returns to scale, the labor share, the investment-capital ratio, and the persistence and dispersion of investment. Financing cost parameters are tied to flotation-cost evidence. The calibration is designed to reproduce broad firm-level investment facts, not to force the cash-flow regression result.

Table 2. Aggregate Results

TABLE 2—AGGREGATE RESULTS	
Variable	Model
Investment share (I/Y)	0.21
Share of financing costs (Λ/Y)	0.0062
External finance/total costs	0.17
Flotation costs/external finance	0.39

Interpretation. The model generates an investment share of 0.21 and external finance equal to 17% of total costs. Although aggregate financing costs are only 0.0062 of output, flotation costs are 39% of external finance. This means outside finance is rare but meaningful when used: high costs discourage small external issues and push firms toward lumpy financing.

Figure 5. Firm Behavior

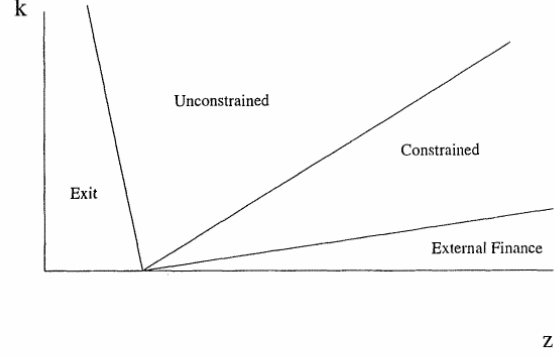


FIGURE 5. FIRM BEHAVIOR

Interpretation. The state-space sketch summarizes the model's policy regions. Low-productivity firms exit; some firms invest only up to internally available funds and are constrained; unconstrained firms invest less than internal funds; and very productive or very small firms pay external-finance costs. The key lesson is that constraint status depends jointly on productivity and capital, not just firm size.

Table 3. Cross-Sectional Results

TABLE 3—CROSS-SECTIONAL RESULTS		
Variable	Data	Model
<i>Matched quantities</i>		
Average size (capital)	80.89	80.89
Investment rate I/K	0.145	0.145
Standard deviation I/K	0.139	0.160
Autocorrelation I/K	0.239	0.191
<i>Other quantities</i>		
Mean q	1.56	1.12
Growth rate (sales)	0.036	0.031
Average CF/K	0.292	0.221
Standard deviation CF/K	0.214	0.091
Negative investment (fraction)	0.08	0.13

Note: CF denotes cash flow

Interpretation. The model matches average capital and the investment rate by construction and approximates investment dispersion and autocorrelation. It also produces average q above one and positive sales growth. The model's cash-flow volatility is lower than in the data, but it still provides a plausible laboratory for studying whether cash flow adds information beyond q .

Table 4. Finance

TABLE 4—FINANCE			
Variable	External finance	Constrained	Unconstrained
Firms	0.07	0.63	0.30
Share of investment	0.79	0.74	-0.53
Size (capital)	55.96	171.93	298.57
Mean I/K	1.20	0.188	-0.086
Tobin's q	1.34	1.14	1.08
Marginal product of capital	0.24	0.22	0.19

Interpretation. Only 7% of firms use external finance, yet these firms account for 79% of investment. The constrained firms are 63% of the population and still invest positively, while unconstrained firms have negative average investment. Firms using external finance have the highest investment rate and marginal product of capital, showing that outside finance is used only when investment opportunities are strong enough to justify the cost.

Table 6. Standard Investment Equations

TABLE 6—STANDARD INVESTMENT EQUATIONS			
Coefficient	Data	Benchmark model	No financing constraints
b_1	0.06 (0.01)	2.82 (0.08)	8.07 (0.10)
$\bar{R}^2$	0.12	0.53	0.84

Note: Standard errors are in parentheses.

Interpretation. The standard investment regression uses q alone. In the data, the regression has a low R^2 of 0.12. The benchmark model generates a higher R^2 of 0.53, while the no-financing-constraint model generates an even higher R^2 of 0.84. The comparison shows that a good fit of investment on q is not itself evidence about financing frictions.

Table 5. Finance and Size

TABLE 5—FINANCE AND SIZE		
Variable	Small firms	Large firms
Fraction of firms	0.72	0.28
Share of investment	0.85	0.15
Size (capital)	139.80	359.79
Mean I/K	0.185	0.035
Fraction constrained	0.61	0.39
Exit rate	0.12	0.01
Tobin's q	1.12	1.15
Marginal product of capital	0.21	0.22

Interpretation. Small firms make up 72% of the population and 85% of investment, and they have a much higher exit rate than large firms. They are more likely to be constrained, but not all small firms are constrained and many can use external finance. This table is central to the paper's warning that size is a noisy proxy for financing constraints.

Table 7. Cash-Flow-Augmented Equations

TABLE 7—CASH-FLOW-AUGMENTED EQUATIONS			
Coefficient	Data	Benchmark model	No financing constraints
b_1	0.06 (0.01)	4.13 (0.39)	11.52 (0.05)
b_2	0.14 (0.04)	-2.67 (0.77)	-10.19 (0.10)
$\bar{R}^2$	0.25	0.53	0.98

Note: Standard errors are in parentheses.

Interpretation. Adding cash flow raises the data R^2 from 0.12 to 0.25, replicating the empirical cash-flow-sensitivity puzzle. In the benchmark model, cash flow does not improve explanatory power beyond q (R^2 stays at 0.53). In the model without financing constraints, cash flow does improve fit strongly. This is the paper's clearest demonstration that cash-flow effects are neither sufficient nor necessary evidence of financing constraints.

Table 8. Standard Investment Equations: Subsamples

Coefficient	Data	Balanced panel	Constrained	Unconstrained	External finance	Large	Small
b_1	0.06 (0.01)	2.12 (0.78)	0.60 (0.02)	1.83 (0.14)	4.79 (0.30)	3.19 (0.07)	1.99 (0.21)
$\bar{R}^2$	0.12	0.58	0.99	0.34	0.98	0.74	0.71

Note: Standard errors are in parentheses.

Interpretation. The q -only regression is run across balanced-panel, constrained, unconstrained, external-finance, large, and small subsamples. The fit varies substantially across groups, reflecting nonlinear policy rules and selection. The constrained and external-finance subsamples have extremely high R^2 values, but this does not imply that cash flow should be a good diagnostic variable.

Table 10. Standard Investment Equations: Alternative Specifications

Coefficient	Data	Financing costs		Marginal q
		Variable only	Fixed only	
b_1	0.06 (0.01)	2.92 (0.07)	2.82 (0.11)	−0.86 (1.03)
$\bar{R}^2$	0.12	0.61	0.32	0.06

Note: Standard errors are in parentheses.

Interpretation. The paper tests alternative financing-cost specifications and a marginal- q specification. The q -only regression remains sensitive to how investment frictions are modeled. The marginal- q column performs particularly poorly, reinforcing the idea that average q can outperform marginal q when investment is lumpy and policy rules are discontinuous.

Table 9. Cash-Flow-Augmented Equations: Subsamples

Coefficient	Data	Balanced panel	Constrained	Unconstrained	External finance	Large	Small
b_1	0.06 (0.01)	8.09 (0.34)	0.04 (0.12)	15.59 (0.94)	12.01 (0.45)	6.15 (0.34)	18.60 (2.20)
b_2	0.14 (0.04)	−12.57 (0.70)	1.13 (0.24)	−22.48 (1.54)	−17.13 (1.00)	−5.82 (0.66)	−37.60 (4.98)
$\bar{R}^2$	0.25	0.61	0.99	0.52	0.99	0.77	0.83

Note: Standard errors are in parentheses.

Interpretation. Adding cash flow does not systematically help in the subsamples where financing constraints are truly present. For constrained firms, R^2 is essentially unchanged. Cash flow adds more fit in the unconstrained and small-firm samples. This is exactly the opposite of what a simple financing-constraint interpretation of cash-flow sensitivity would predict.

Table 11. Cash-Flow-Augmented Equations: Alternative Specifications

Coefficient	Data	Financing costs		Marginal q
		Variable only	Fixed only	
b_1	0.06 (0.01)	4.88 (0.29)	6.25 (0.50)	−0.22 (0.45)
b_2	0.14 (0.04)	−4.11 (0.59)	−7.00 (1.00)	5.41 (0.16)
$\bar{R}^2$	0.25	0.63	0.35	0.50

Note: Standard errors are in parentheses.

Interpretation. Cash flow adds little in the two alternative financing-cost cases, but it adds substantial explanatory power when marginal q is used. The interpretation is not that cash flow reveals financing constraints; rather, cash flow helps compensate for a poor measure of fundamentals.

Table 12. Cross Correlations: Benchmark Model

TABLE 12—CROSS CORRELATIONS:
BENCHMARK MODEL

Variable	Investment	Tobin's q	Cash flow	Sales	exp(z)
Investment	1.00	0.53	0.57	0.56	0.53
Tobin's q		1.00	0.92	0.93	0.92
Cash flow			1.00	0.99	0.96
Sales				1.00	0.95
exp(z)					1.00

Interpretation. Investment, Tobin's q , cash flow, sales, and productivity are all strongly correlated in the benchmark model. Cash flow is 0.92 correlated with q and 0.96 correlated with productivity. This high collinearity makes it difficult for a linear regression to assign economic meaning to the cash-flow coefficient.

Table 14. Standard Investment Equations: Measurement Error

TABLE 14—STANDARD INVESTMENT EQUATIONS: MEASUREMENT ERROR

Coefficient	Data	Measurement error Capital stock	Measurement error Classical	Measurement error No constraints
b_1	0.06 (0.01)	2.08 (0.24)	1.59 (0.08)	6.71 (0.12)
R^2	0.12	0.29	0.18	0.73

Note: Standard errors are in parentheses.

Interpretation. Introducing measurement error weakens the explanatory power of the q -only regression. Measurement error in the capital stock and classical noise in q lower the R^2 relative to the cleaner benchmark. This sets up the key result that cash flow can look useful because it proxies for fundamentals that noisy q fails to capture.

Table 13. Cross Correlations: No Financing Constraints

TABLE 13—CROSS CORRELATIONS:
NO FINANCING CONSTRAINTS

Variable	Investment	Tobin's q	Cash flow	Sales	exp(z)
Investment	1.00	0.45	0.45	0.48	0.42
Tobin's q		1.00	0.95	0.97	0.95
Cash flow			1.00	0.99	0.98
Sales				1.00	0.97
exp(z)					1.00

Interpretation. The same strong collinearity appears even when financing constraints are removed. Cash flow remains highly correlated with q , sales, and productivity. Therefore, a cash-flow coefficient can be generated by productivity-driven investment opportunities rather than by financial frictions.

Table 15. Cash-Flow-Augmented Equations: Measurement Error

TABLE 15—CASH-FLOW-AUGMENTED EQUATIONS: MEASUREMENT ERROR

Coefficient	Data	Measurement error Capital stock	Measurement error Classical	Measurement error No constraints
b_1	0.06 (0.01)	0.65 (0.61)	0.45 (0.09)	8.19 (0.14)
b_2	0.14 (0.04)	1.25 (0.40)	4.72 (0.22)	-5.35 (0.32)
R^2	0.25	0.39	0.46	0.81

Note: Standard errors are in parentheses.

Interpretation. When q is measured with error, adding cash flow raises explanatory power. In the no-constraints case, R^2 rises from 0.73 to 0.81 after adding cash flow. The result shows that cash-flow sensitivity can be an artifact of measurement error and misspecification even when no financing frictions exist.

6. Takeaways for Paper Reading

- Do not interpret a significant investment-cash flow coefficient as direct evidence of binding financing constraints.
- If financing frictions affect firm value, a correct measure of q should already incorporate them.
- Subsample splits by size or dividend payout can be useful descriptively, but they do not solve the identification problem.
- Measurement error in q and omitted productivity shocks can make cash flow look important even when it is not a causal source of investment financing.
- The paper is best read as a warning about reduced-form investment regressions and as an argument for structural discipline when interpreting cash-flow sensitivity.

7. Possible Limitations and Open Questions

- The model abstracts from some financial choices, such as persistent cash holdings, heterogeneous debt contracts, and rich intermediary behavior.
- The calibration is designed to match broad investment moments, but some model-generated moments, such as average q and cash-flow volatility, differ from the data.
- The paper shows what cash-flow regressions can and cannot prove; it does not claim that financing constraints are unimportant in reality.
- Future work can use richer structural models with aggregate shocks, monetary policy, and explicit balance-sheet dynamics to quantify the macroeconomic importance of financing frictions.